

<pre> 1. use ProgBooksDB db.createCollection("ProgrammingBooks") db.ProgrammingBooks.insertOne({ title: "The Pragmatic Programmer: Your Journey to Mastery", author: "David Thomas, Andrew Hunt", category: "Software Development", year: 1999 }) db.ProgrammingBooks.insertMany([{ title: "Clean Code: A Handbook of Agile Software Craftsmanship", author: "Robert C. Martin", category: "Software Development", year: 2008 }, { title: "JavaScript: The Good Parts", author: "Douglas Crockford", category: "JavaScript", year: 2008 }, { title: "Design Patterns: Elements of Reusable Object-Oriented Software", author: "Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides", category: "Software Design", year: 1994 }, { title: "Introduction to Algorithms", author: "Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein", category: "Algorithms", year: 1990 }, { title: "Python Crash Course: A Hands-On, Project-Based Introduction to Programming", author: "Eric Matthes", category: "Python", year: 2015 }]) db.ProgrammingBooks.find().pretty() db.ProgrammingBooks.find({ year: { \$gt: 2000 } }).pretty() db.ProgrammingBooks.updateOne({ title: "Clean Code: A Handbook of Agile Software Craftsmanship" }, { \$set: { author: "Robert C. Martin (Uncle Bob)" } }) </pre>	<pre> db.ProgrammingBooks.find({ year: { \$eq: 2008 } }).pretty() db.ProgrammingBooks.find({ author: { \$regex: "Robert*" } }).pretty() db.ProgrammingBooks.updateMany({ year: { \$lt: 2010 } }, { \$set: { category: "Classic Programming Books" } }) db.ProgrammingBooks.find({ year: { \$lt: 2010 } }).pretty() db.ProgrammingBooks.deleteOne({ title: "JavaScript: The Good Parts" }) db.ProgrammingBooks.find({ title: "JavaScript: The Good Parts" }).pretty() </pre> <pre> 2. use MoviesDB db.createCollection("Movies") db.Movies.insertMany([{ title: "Inception", director: "Christopher Nolan", genre: "Science Fiction", year: 2010, ratings: { imdb: 8.8, rottenTomatoes: 87 } }, { title: "The Matrix", director: "Wachowskis", genre: "Science Fiction", year: 1999, ratings: { imdb: 8.7, rottenTomatoes: 87 } }, { title: "The Godfather", director: "Francis Ford Coppola", genre: "Crime", year: 1972, ratings: { imdb: 9.2, rottenTomatoes: 97 } }, { title: "Pulp Fiction", director: "Quentin Tarantino", genre: "Crime", year: 1994, ratings: { imdb: 8.9, rottenTomatoes: 92 } }, { title: "The Shawshank Redemption", director: "Frank Darabont", genre: "Drama", year: 1994, ratings: { imdb: 9.3, rottenTomatoes: 91 } }, { title: "The Dark Knight", director: "Christopher Nolan", genre: "Action", year: 2008, ratings: { imdb: 9.0, rottenTomatoes: 94 } }, { title: "Fight Club", director: "David Fincher", genre: "Drama", year: 1999, ratings: { imdb: 8.8, rottenTomatoes: 79 } }]) db.Movies.find({}, { title: 1, director: 1 }) db.Movies.find({}, { title: 1, director: 1, _id: 0 }) db.Movies.find({}, { ratings: 0 }) db.Movies.find({ director: "Christopher Nolan" }, { title: 1, year: 1, _id: 0 }) db.Movies.find({}, { title: 1, director: 1, year: 1, _id: 0 }).limit(5) </pre>
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4. use retailDB

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db.Products.insertMany([
  {
    name: "Laptop",
    brand: "BrandA",
    features: [
      { name: "Processor", value: "Intel i7" },
      { name: "RAM", value: "16GB" },
      { name: "Storage", value: "512GB SSD" }
    ],
    reviews: [
      { user: "Alice", rating: 5, comment: "Excellent!" },
      { user: "Bob", rating: 4, comment: "Very good" },
      { user: "Charlie", rating: 3, comment: "Average" }
    ]
  },
  {
    name: "Smartphone",
    brand: "BrandB",
    features: [
      { name: "Processor", value: "Snapdragon 888" },
      { name: "RAM", value: "8GB" },
      { name: "Storage", value: "256GB" }
    ],
    reviews: [
      { user: "Dave", rating: 4, comment: "Good phone" },
      { user: "Eve", rating: 2, comment: "Not satisfied" }
    ]
  }
])
db.Products.find(
  { name: "Laptop", "reviews.user": "Alice" },
  { "reviews.$": 1 }
).pretty()
db.Products.find(
  { name: "Laptop" },
  { reviews: { $elemMatch: { rating: { $gt: 4 } } } }
).pretty()
db.Products.find(
  { name: "Smartphone" },
  { reviews: { $slice: 1 } }
).pretty()
```

5. use salesDB

```
db.Sales.insertMany([
  { date: new Date("2024-01-01"), product: "Laptop", price: 1200, quantity: 1, customer: "Amar" },
  { date: new Date("2024-01-02"), product: "Laptop", price: 1200, quantity: 2, customer: "Babu" },
  { date: new Date("2024-01-03"), product: "Mouse", price: 25, quantity: 5, customer: "Chandra" },
  { date: new Date("2024-01-04"), product: "Keyboard", price: 45, quantity: 3, customer: "Amar" },
  { date: new Date("2024-01-05"), product: "Monitor", price: 300, quantity: 1, customer: "Babu" },
  { date: new Date("2024-01-06"), product: "Laptop", price: 1200, quantity: 1, customer: "Deva" }
])
db.Sales.aggregate([
  {
    $group: {
      _id: "$product",
      averagePrice: { $avg: "$price" } } }
]).pretty()
db.Sales.aggregate([
  {
    $group: {
      _id: "$product",
      minPrice: { $min: "$price" } } } ]).pretty()
db.Sales.aggregate([
  {
    $group: {
      _id: "$product",
      maxPrice: { $max: "$price" } } } ]).pretty()
db.Sales.aggregate([
  {
    $group: {
      _id: "$customer",
      products: { $push: "$product" } } } ]).pretty()
db.Sales.aggregate([
  {
    $group: {
      _id: "$customer",
      uniqueProducts: { $addToSet: "$product" } } }
]).pretty()
```

3. use companyDB <pre> db.Employees.insertMany([{ name: "Alice", age: 30, department: "HR", salary: 50000, joinDate: new Date("2015-01-15") }, { name: "Bob", age: 24, department: "Engineering", salary: 70000, joinDate: new Date("2019-03-10") }, { name: "Charlie", age: 29, department: "Engineering", salary: 75000, joinDate: new Date("2017-06-23") }, { name: "David", age: 35, department: "Marketing", salary: 60000, joinDate: new Date("2014-11-01") }, { name: "Eve", age: 28, department: "Finance", salary: 80000, joinDate: new Date("2018-08-19") }]) db.Employees.find({ department: { \$eq: "Engineering" } }).pretty() db.Employees.find({ department: { \$ne: "HR" } }).pretty() db.Employees.find({ age: { \$gt: 30 } }).pretty() db.Employees.find({ salary: { \$lt: 70000 } }).pretty() db.Employees.find({ joinDate: { \$gte: new Date("2018-01-01") } }).pretty() db.Employees.find({ age: { \$lte: 28 } }).pretty() db.Employees.find({ \$and: [{ department: "Engineering" }, { salary: { \$gt: 70000 } }] }).pretty() db.Employees.find({ \$or: [{ department: "HR" }, { salary: { \$lt: 60000 } }] }).pretty() db.Employees.find({ department: { \$not: { \$eq: "Engineering" } } }).pretty() db.Employees.find({ \$nor: [{ department: "HR" }, { salary: { \$gt: 75000 } }] }).pretty() use geoDatabase db.Places.insertMany([{ name: "Central Park", location: { type: "Point", coordinates: [-73.9654, 40.7829] } }, { name: "Times Square", location: { type: "Point", coordinates: [-73.9851, 40.7580] } }, { name: "Brooklyn Bridge", location: { type: "Point", coordinates: [-73.9969, 40.7061] } }, { name: "Empire State Building", location: { type: "Point", coordinates: [-73.9857, 40.7488] } }, { name: "Statue of Liberty", location: { type: "Point", coordinates: [-74.0445, 40.6892] } }]) db.Places.createIndex({ location: "2dsphere" }) </pre>	<pre> db.Places.find({ location: { \$near: { \$geometry: { type: "Point", coordinates: [- 73.9851, 40.7580] }, \$maxDistance: 5000 } } }).pretty() db.Places.find({ location: { \$geoWithin: { \$geometry: { type: "Polygon", coordinates: [[[-70.016, 35.715], [-74.014, 40.717], [-73.990, 40.730], [-73.990, 40.715], [-70.016, 35.715]]] } } } }).pretty() use techDB db.Devices.insertMany([{ name: "Device A", status: 5 }, { name: "Device B", status: 3 }, { name: "Device C", status: 12 }, { name: "Device D", status: 10 }, { name: "Device E", status: 7 }]) db.Devices.find({ status: { \$bitsAllSet: [0, 2] } }).pretty() db.Devices.find({ status: { \$bitsAnySet: [1] } }).pretty() db.Devices.find({ status: { \$bitsAllClear: [1, 3] } }).pretty() db.Devices.find({ status: { \$bitsAnyClear: [0] } }).pretty() </pre>
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